

Appendix: scoped morphological view

Dataset `loss_mitigation_forms` **Component** `risk-pooling` **Mechanism** `MC1 = pooling`
29 forms $\times$ 9 characteristics. This view is restricted to one declared component set. Comparative claims never run on the full characteristic set: with enough characteristics any two forms separate, so a matrix printed over all of them reports how many characteristics were used rather than anything about the forms. At this n the matrix below is a coverage map and not evidence; the comparative reading is confined to Table 2, whose two characteristics were fixed before the coding.

Table 1: Scoped matrix. States are coded; see the character-state list. Missingness is never collapsed: - no row entered, .NR not recorded in the source, .IL illegible, .NA inapplicable, 0 an observed absence.

form	CN1	HZ1	HZ2	MB3	MC1	PR1	PY0	PY1	PY2
averia_pool	0	1	1	1	2	0	0	.NA	.NA
confraternity_fund_it	0	2	1	0	2	0	1	0	.NR
craft_pension_edinburgh	0	.NR	-	1	2	0	1	3	4
craft_widows_fund_edinburgh	0	.NR	-	0	2	P	1	1	.NR
friendly_society_england	0	.NR	P	0	2	0	1	0	3
friendly_society_female_england	0	.NR	P	0	2	P	1	0	3
general_average	2	2	P	1	2	0	1	0	3
guild_box_brabant	0	.NR	-	1	2	0	1	0	3
guild_box_dutch	0	2	P	1	2	0	1	0	3
guild_relief_england	.NR	.NR	-	0	2	0	1	3	4
ko_daimanin_1828	-	-	-	-	2	-	-	-	.NR
ko_furuichi_sengan	0	.NR	-	1	2	0	2	2	.NR
ko_hokinosawa_1820	0	.NR	-	0	2	0	2	2	0
ko_hokoin_1832	0	.NR	-	0	2	0	2	2	.NR
ko_horyuji_50kan	0	.NR	-	.NR	2	0	2	2	.NR
ko_kosaburo_1802	0	.NR	-	.NR	2	0	2	2	.NR
ko_meio_1498	0	.NR	-	.NR	2	0	2	2	.NR
ko_mitarai_gin_1805	0	.NR	-	0	2	0	2	2	.NR
ko_mitarai_hanei_1840	0	.NR	-	0	2	0	2	2	2
ko_mitarai_mujin_1773	0	.NR	-	0	2	0	2	2	0
ko_mochiyori_yutsu_1876	0	.NR	-	0	2	0	2	2	2
ko_shijunin_1829	0	-	-	-	2	-	-	-	0
ko_usui_bango_1822	0	.NR	-	0	2	0	2	2	.NR
ko_yamakuni_1555	0	.NR	-	.NR	2	0	2	2	.NR
tontine	-	-	-	-	2	.NR	-	-	-
tontine_en_1693	1	0	-	0	2	0	2	1	.NR
tontine_fr_royal	-	0	-	0	2	P	2	-	-
torinoke_mujin	0	.NR	-	0	2	0	2	2	0
widows_fund_scotland	-	0	-	1	2	1	1	1	-

Character states

CN1 *contribution timing* — 0 = ex-ante-periodic; 1 = ex-ante-lump; 2 = ex-post-assessment

HZ1 *hazard correlation* — 0 = idiosyncratic; 1 = covariate; 2 = mixed

HZ2 *hazard responsiveness* — ternary: 1 present, P partial, 0 absent

MB3 *participation basis* — 0 = voluntary; 1 = compulsory

Table 2: The comparative reading, printed with full values and no state codes. PR1 takes both values *inside* MC1=pooling at a fixed date, so ex-ante pricing is a separate axis rather than a stage of mechanism; PY0 separates the forms on whether a draw corresponds to the recipient's own realised loss.

form	PR1 peril priced ex ante	PY0 pool output
averia_pool	0	collective-good
confraternity_fund_it	0	individual-indemnity
craft_pension_edinburgh	0	individual-indemnity
craft_widows_fund_edinburgh	P	individual-indemnity
friendly_society_england	0	individual-indemnity
friendly_society_female_england	P	individual-indemnity
general_average	0	individual-indemnity
guild_box_brabant	0	individual-indemnity
guild_box_dutch	0	individual-indemnity
guild_relief_england	0	individual-indemnity
ko_daimanin_1828	-	-
ko_furuichi_sengan	0	individual-draw
ko_hokinosawa_1820	0	individual-draw
ko_hokoin_1832	0	individual-draw
ko_horyuji_50kan	0	individual-draw
ko_kosaburo_1802	0	individual-draw
ko_meio_1498	0	individual-draw
ko_mitarai_gin_1805	0	individual-draw
ko_mitarai_hanei_1840	0	individual-draw
ko_mitarai_mujin_1773	0	individual-draw
ko_mochiyori_yutsu_1876	0	individual-draw
ko_shijunin_1829	-	-
ko_usui_bango_1822	0	individual-draw
ko_yamakuni_1555	0	individual-draw
tontine	.NR	-
tontine_en_1693	0	individual-draw
tontine_fr_royal	P	individual-draw
torinoke_mujin	0	individual-draw
widows_fund_scotland	1	individual-indemnity

MC1 *mitigation mechanism* — 0 = allocation; 1 = spreading; 2 = pooling
 PR1 *peril priced ex ante* — ternary: 1 present, P partial, 0 absent
 PY0 *pool output* — 0 = collective-good; 1 = individual-indemnity; 2 = individual-draw
 PY1 *payout trigger* — 0 = realised-loss; 1 = life-event; 2 = rotation; 3 = need-assessed
 PY2 *allocation rule* — 0 = rotation-lot; 1 = rotation-fixed; 2 = auction; 3 = indemnity; 4 = need-assessed

Forms

averia_pool Avería (compulsory convoy levy) *Spanish / Carrera de Indias, 1521-1681*
confraternity_fund_it Confraternity and guild mutual support - Italian *Latin Christendom / Italian, 11c-15c*
craft_pension_edinburgh Craft pensions and supply, incorporation of Mary's Chapel *Scottish / Edinburgh, 1670-1768*
craft_widows_fund_edinburgh Widows' fund annuity scheme, incorporation of Mary's Chapel *Scottish / Edinburgh, 1768-*
friendly_society_england Friendly society / box club *English, 1870-1914 (this coding)*
friendly_society_female_england Female friendly society *English, 1780-1830 (this coding)*
general_average General average (jettison contribution) *Mediterranean / lex Rhodia, antiquity onward*
guild_box_brabant Ambachtsbus / armenbus — craft guild sickness box *Low Countries / Brabant, 1250-1600*
guild_box_dutch Gildebus — guild sickness, burial, old-age and widows fund *Dutch Republic, 1550-1800*
guild_relief_england English religious gild relief *English, c.1350-1400*
ko_daimanin_1828 ??? (???) *Japanese / Kawachi, 1828-1833+*
ko_furuichi_sengan ????? (??, ? ?????) *Japanese / Yamato, late 15c*
ko_hokinosawa_1820 ???????? (?????, ???) *Japanese / Echigo, 1820-1838*
ko_hokoin_1832 ????????? (the kō joined by Hōkō-in) *Japanese / Dewa-Yamagata, 1832-*
ko_horyuji_50kan ????????? *Japanese / Yamato, Muromachi (undated)*
ko_kosaburo_1802 ????? (?????) *Japanese / Echigo, 1802-*
ko_meio_1498 ??? ????? (?????) *Japanese / Kii, 1498*
ko_mitarai_gin_1805 ?????(????) *Japanese / Aki, 1805-*
ko_mitarai_hanei_1840 ?????(????) *Japanese / Aki, 1840-*
ko_mitarai_mujin_1773 ???(???, ????? ?????) *Japanese / Aki, 1773-1785*
ko_mochiyori_yutsu_1876 ????? (Mochiyori Yūtsū-sha) *Japanese / Kawachi-Settsu, 1876-1884*
ko_shijunin_1829 ??? (???) *Japanese / Kawachi, 1829-1843*
ko_usui_bango_1822 ????? (???) *Japanese / Kawachi, 1822-*
ko_yamakuni_1555 ????? (?????, ? ?????) *Japanese / Tanba, 1555-1563*

tontine	Tontine	<i>European, 17c onward</i>
tontine_en_1693	Tontine — English (Million Act 5 & 6 Will. & Mar. c. 5)	<i>English, 1693-1789</i>
tontine_fr_royal	Tontine — French royal series (fourteen age classes)	<i>French, 1689-1759</i>
torinoke_mujin	????? (winner released from further contribution)	<i>Japanese, 17c-1930s</i>
widows_fund_scotland	Scottish Ministers' Widows' Fund	<i>Scottish, 1744 onward</i>

All codings marked **coder=ai** are provisional and pending human verification. Cells carrying **[verify]** in **source_ref** are asserted against an unpinned citation.